

ServiceNow Platform | Academic Project

SMART LIBRARY

REQUEST WORKFLOW

Automating Library Services using ServiceNow

TEAM MEMBERS

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1. INTRODUCTION

1.1 Project Title

Smart Library Request Workflow

A ServiceNow-Powered Full Stack Library Automation System

1.2 Team Members

#	Name	Role	Responsibility
1	SANTHOSHKUMAR. R	Team Lead / Backend Dev	ServiceNow Flow Designer, API integration, Workflow engine
2	DURGA. R	Frontend Developer	Service Portal UI, Widget development, AngularJS components
3	MADHUSHINI. R	Database & Integration	Custom tables, CMDB, LMS REST API integration, MID Server
4	ROHINI. M	Testing & Documentation	ATF test suites, performance testing, technical documentation

1.3 Technology Stack Overview

Layer	Technology	Version	Notes
Platform	ServiceNow	Utah Release	Core ITSM + custom scoped app
Frontend	Service Portal (AngularJS)	1.8.x	Responsive self-service UI
Backend	Flow Designer + Business Rules	ServiceNow	Workflow automation engine
Database	ServiceNow Tables + CMDB	Cloud	Relational + configuration data
Integration	REST API / MID Server	v2	LMS, LDAP, Twilio (SMS)
Auth	SAML 2.0 / LDAP SSO	2.0	Institutional single sign-on
Testing	ATF + Apache JMeter	5.5	Automated + performance testing
Reporting	Performance Analytics (PA)	ServiceNow	KPI dashboards

2. PROJECT OVERVIEW

2.1 Purpose

The **Smart Library Request Workflow** is a full-stack application built entirely on the ServiceNow platform. It digitises and automates all library service interactions — from book discovery and reservation to inter-library loans, renewal management, and overdue tracking — providing a seamless, 24/7 self-service experience for students and faculty while reducing librarian workload through intelligent automation.

2.2 Key Features

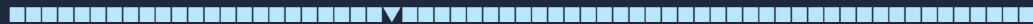
■ Real-Time Catalog Search	Live integration with the Library Management System (LMS) for instant book availability.
■ Multi-Type Request Workflow	Separate workflows for New Book Requests, Renewals, and Inter-Library Loans (ILL).
■ ■ Automated Approvals	Condition-based auto-approval for renewals; role-based manual approval for ILL requests.
■ SLA & Escalation Engine	Automated SLA tracking with email/SMS alerts and manager escalation on breach.
■ Role-Based Access Control	Separate dashboards and permissions for Students, Faculty, Librarians, and Admins.
■ Analytics Dashboard	Performance Analytics widgets showing request volumes, SLA compliance, and trends.
■ Omnichannel Notifications	Email (SMTP) and SMS (Twilio) alerts at every workflow state transition.
■ Mobile-Responsive Portal	AngularJS Service Portal accessible from any device — desktop, tablet, or mobile.

2.3 Target Users

User Role	Access Level	Primary Actions
Student	Portal User	Search catalog, submit requests, track status, renew books
Faculty	Portal User + Priority Queue	Submit ILL, request bulk reservations, access faculty shelf
Librarian	Fulfillment Agent	Process tasks, manage inventory, update catalog, override SLA
Library Manager	Manager Dashboard	Approve ILL, view reports, configure SLAs, manage staff
System Admin	Full Admin	Configure workflows, manage users, maintain integrations



■ HTTPS / REST



■ APPLICATION LAYER ■

■ Flow Designer | Business Rules | Script Includes | ATF ■

■ Scripted REST APIs | Notification Engine | SLA Engine ■



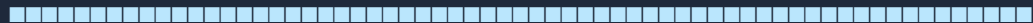
■ GlideRecord / GlideQuery



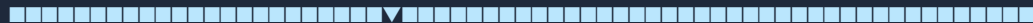
■ DATA LAYER ■

■ u_library_catalog | u_borrow_record | u_library_request ■

■ sc_request | task | sys_user | cmdb tables ■

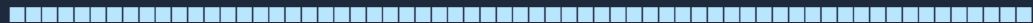


■ MID Server / REST



■ INTEGRATION LAYER ■

■ LMS (Koha) | LDAP/AD | Twilio SMS | SMTP Email ■



4. SETUP INSTRUCTIONS

4.1 Prerequisites

Requirement	Version / Detail	Purpose
ServiceNow Instance	Utah / Vancouver release	Application hosting
Admin Access	System Administrator role	Full configuration rights
MID Server	Latest compatible version	On-premise LMS bridge
Node.js (for local tooling)	18.x LTS	ServiceNow CLI / dev utilities
ServiceNow CLI (snc)	Latest	Update set deployment
Git	2.40+	Source code version control
Apache JMeter	5.5	Performance testing
Postman	Latest	API endpoint testing
LDAP Server	Active Directory / OpenLDAP	User authentication
Twilio Account	API credentials	SMS notification dispatch

4.2 Installation Steps

Step 1: Clone Repository & Prepare Update Set

```
git clone https://github.com/your-org/smart-library-servicenow.git
cd smart-library-servicenow
# Locate: /update_sets/SmartLibrary_v1.0.xml
```

Step 2: Import Update Set into ServiceNow

```
# Navigate to: System Update Sets > Retrieved Update Sets
# Click 'Import Update Set from XML'
# Upload: SmartLibrary_v1.0.xml
# Preview → Accept all > Commit Update Set
```

Step 3: Configure LDAP Integration

```
# Navigate to: LDAP > LDAP Servers
# Create new LDAP server record:
# Name: Institutional_LDAP
# Server URL: ldap://ldap.yourinstitution.edu:389
# Base DN: ou=users,dc=yourinstitution,dc=edu
# Test connection → Save
```

Step 4: Configure MID Server for LMS Integration

```
# Deploy MID Server on LMS host machine
# Edit config.xml: set servicenow instance URL + credentials
# Navigate to: MID Server > Servers → validate status = UP
# Configure JDBC connection to Koha LMS database
```

Step 5: Set Twilio SMS Credentials

```
# Navigate to: System Properties > Smart Library > SMS Config
# Set: sl.sms.account_sid =
# Set: sl.sms.auth_token =
# Set: sl.sms.from_number = +1XXXXXXXXXX
```

Step 6: Activate the Service Portal

```
# Navigate to: Service Portal > Portals
# Find: Smart Library Portal (sl_portal)
# Set as default or note portal suffix: /library_portal
# Verify portal loads at: https://.service-now.com/library_portal
```

5. FOLDER STRUCTURE

5.1 Repository Root Structure

```
smart-library-servicenow/
├── update_sets/ # Exportable ServiceNow update sets
│   └── SmartLibrary_v1.0.xml
├── client/ # Frontend (Service Portal) assets
├── server/ # Backend scripts & configurations
├── tests/ # ATF test suites + JMeter plans
├── docs/ # Project documentation
├── integration/ # MID Server configs & REST schemas
└── README.md
```

5.2 Client (Frontend) Directory

```
client/
├── portal/
│   ├── sl_portal.json # Portal configuration export
│   └── theme/
│       ├── sl_theme.scss # Custom SCSS styles
│       ├── sl_variables.scss # Color tokens & typography
│       └── pages/
│           ├── home.json # Home page config
│           ├── my_requests.json # My Requests page
│           ├── catalog.json # Book catalog page
│           └── request_detail.json # Request detail page
├── widgets/
│   ├── sl-catalog-search/ # Live catalog search widget
│   ├── body.html
│   ├── client_script.js
│   ├── server_script.js
│   └── option_schema.json
├── sl-request-form/ # Book request submission widget
├── sl-my-requests/ # Request tracker widget
├── sl-notifications/ # In-portal notification banner
└── sl-availability-badge/ # Real-time availability indicator
```

5.3 Server (Backend) Directory

```
server/
├── flows/
│   ├── book_request_flow.xml # Main book request Flow Designer export
│   ├── renewal_flow.xml # Renewal workflow
│   └── ill_request_flow.xml # Inter-library loan workflow
└── business_rules/
```

```
■ ■■■ br_validate_request.js # Pre-insert validation
■ ■■■ br_set_sla_target.js # SLA assignment on submission
■ ■■■ br_catalog_sync_trigger.js # Trigger catalog availability update
■■■ script_includes/
■ ■■■ LibraryCatalogUtil.js # Catalog API helper class
■ ■■■ SLACalculator.js # Business-hours SLA computation
■ ■■■ NotificationHelper.js # Email + SMS dispatch utility
■ ■■■ ILLManager.js # Inter-library loan logic
■■■ scripted_rest/
■ ■■■ catalog_api.js # GET /api/sl/catalog/{isbn}
■ ■■■ request_api.js # POST/GET /api/sl/request
■ ■■■ availability_api.js # GET /api/sl/availability
■■■ scheduled_jobs/
■ ■■■ sj_overdue_check.js # Daily overdue sweep
■ ■■■ sj_catalog_sync.js # Nightly LMS catalog sync
■■■ tables/
■■■ schema.xml # Custom table definitions export
```

6. RUNNING THE APPLICATION

6.1 Accessing the Frontend Portal

```
# Production URL (after deployment):
https://.service-now.com/library_portal

# Local development using ServiceNow CLI:
npm install -g @servicenow/cli
snc login --url https://.service-now.com
snc ui-component develop # Hot-reload widget development

# Sync portal changes to instance:
snc ui-component deploy --force
```

6.2 Starting the Backend Workflows

```
# Activate Flow Designer flows:
# Navigate to: Process Automation > Flow Designer
# Open each flow > Click 'Activate'
# Flows to activate:
# - Smart Library - Book Request Flow
# - Smart Library - Renewal Flow
# - Smart Library - ILL Request Flow

# Verify Business Rules are active:
# Navigate to: System Definition > Business Rules
# Filter by: Application = Smart Library
# Ensure all rules show Active = true

# Start Scheduled Jobs:
# Navigate to: System Definition > Scheduled Jobs
# Run 'SL - Catalog Sync' to verify LMS connection
# Run 'SL - Overdue Check' to verify SLA engine
```

6.3 MID Server Health Check

```
# On MID Server host machine:
cd /opt/servicenow/mid/agent
./mid.sh status # Check MID Server status
./mid.sh start # Start if not running

# Verify in ServiceNow:
# Navigate to: MID Server > Servers
# Status should be: Validated
# Run: Validate > Test connectivity to LMS
```

7. API DOCUMENTATION

Base URL: <https://<instance>.service-now.com/api/sl/v1>

All endpoints require Bearer Token (OAuth 2.0) or Basic Auth header.

7.1 Catalog Endpoints

GET	/catalog/search?q={query}	Search catalog by title, author, or subject keyword
GET	/catalog/{isbn}	Get book details and real-time availability count
GET	/catalog/available	List all currently available books
POST	/catalog/reserve	Reserve a specific book for a user

Sample Request – GET /catalog/{isbn}

```
GET /api/sl/v1/catalog/9780134685991 HTTP/1.1
Host: .service-now.com
Authorization: Bearer
Content-Type: application/json
```

Sample Response

```
{
  "result": {
    "sys_id": "a1b2c3d4e5f6...",
    "title": "Clean Code: A Handbook of Agile Software Craftsmanship",
    "author": "Robert C. Martin",
    "isbn": "9780134685991",
    "available_qty": 3,
    "total_qty": 5,
    "location": "Shelf B-12, Floor 2",
    "category": "Software Engineering"
  }
}
```

7.2 Request Management Endpoints

POST	/request/book	Submit a new book borrow request
POST	/request/renewal	Request a loan extension for borrowed book
POST	/request/ill	Submit an inter-library loan request

GET	/request/{request_id}	Get status and details of a specific request
GET	/request/user/{user_id}	List all requests for a specific user
PUT	/request/{request_id}/cancel	Cancel a pending request

Sample Request – POST /request/book

```
POST /api/sl/v1/request/book HTTP/1.1
Authorization: Bearer
Content-Type: application/json

{
  "book_isbn": "9780134685991",
  "requested_by": "student.user@institution.edu",
  "pickup_location": "Main Library Counter",
  "preferred_date": "2025-08-20",
  "notes": "Required for final year project"
}
```

7.3 Notification & SLA Endpoints

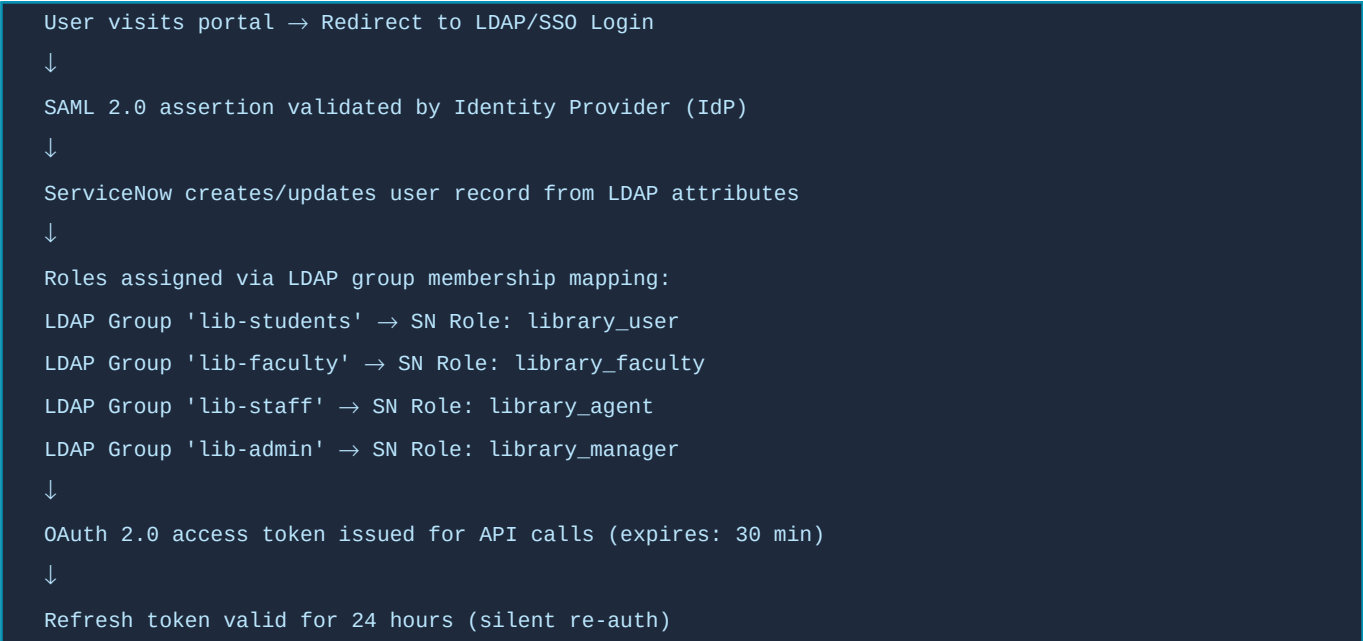
GET	/sla/active	List requests currently within SLA window
GET	/sla/breached	List requests that have breached SLA
POST	/notification/send	Manually trigger notification for a request
GET	/notification/log/{user_id}	Get notification history for a user

7.4 Error Codes

HTTP Code	Error Code	Message	Resolution
400	SL-001	Invalid ISBN format	Provide 13-digit ISBN without hyphens
401	SL-002	Authentication failed	Refresh OAuth token or re-login
403	SL-003	Insufficient privileges	Ensure user has correct role assigned
404	SL-004	Book not found in catalog	Verify ISBN or search by title
409	SL-005	Book already reserved by user	Cancel existing reservation first
429	SL-006	Rate limit exceeded	Max 100 requests/min per user
500	SL-500	Internal server error	Contact system administrator

8. AUTHENTICATION

8.1 Authentication Flow



8.2 Role-Based Access Control (RBAC)

Role	ServiceNow Role Name	Portal Access	API Scope
Student	library_user	Home, Search, My Requests, Renewal	/catalog, /request (own only)
Faculty	library_faculty	All student + Priority ILL queue	/catalog, /request, /ill
Librarian	library_agent	Fulfillment dashboard, Inventory	/catalog (write), /request (all)
Manager	library_manager	Full portal + Analytics	/catalog, /request, /sla, /reports
Admin	admin	System configuration	Full API access

8.3 Session & Token Management

- Session timeout: 30 minutes of inactivity (configurable via System Property: glide.ui.session_timeout).
- OAuth tokens stored in browser memory only — not in localStorage/sessionStorage.
- Token refresh handled transparently by Angular HTTP interceptor in portal widgets.
- All API traffic encrypted via TLS 1.3; certificates auto-managed by ServiceNow SaaS.
- Failed login attempts: account locked after 5 consecutive failures; auto-unlock in 15 minutes.
- Audit log: All authentication events recorded in sys_audit_delete and syslog tables.

9. USER INTERFACE

9.1 Portal Screens & Components

Home Dashboard

- Prominent search bar with autocomplete (triggers catalog API on 3+ chars).
- Category cards: New Request | My Active Borrows | Renewals Due | ILL Requests.
- Alert banner showing overdue items (red) and items ready for pickup (green).
- Quick-stats row: Active Borrows, Due This Week, Pending Requests.

Catalog Search & Book Detail

- Grid/list toggle view of search results with cover art placeholder, availability badge.
- Book Detail modal: full metadata, availability count, shelf location, borrow history.
- One-click 'Request This Book' CTA button opening inline request form.
- Waitlist indicator with estimated availability date for unavailable books.

Request Submission Form

- Dynamic form adapts fields based on request type (New / Renewal / ILL).
- Real-time field validation with inline error messages.
- Pickup location selector with building map tooltip.
- Confirmation dialog with request number and expected fulfillment date.

My Requests Tracker

- Timeline view of all requests with color-coded status indicators.
- Expandable row for each request showing full workflow history.
- Inline renewal button (enabled only when eligible).
- Download request history as CSV for academic record-keeping.

Librarian Fulfillment Dashboard

- Queue table sorted by SLA remaining time with traffic-light color coding.
- Bulk action toolbar: Assign, Complete, Escalate, Send Reminder.
- Live SLA breach counter widget at top of dashboard.
- Filter by request type, assignment group, and date range.

9.2 UI Design System

Token	Value	Usage
Primary Color	#1D4ED8 (Blue)	Buttons, links, active states

Token	Value	Usage
Secondary Color	#0891B2 (Cyan)	Accents, badges, hover states
Background	#FFF6FF	Card backgrounds, info panels
Text Primary	#1E293B	Headings, body text
Text Muted	#64748B	Labels, captions, placeholders
Success	#059669 (Green)	Available status, success alerts
Warning	#D97706 (Amber)	Due soon, partial availability
Error	#DC2626 (Red)	Overdue, unavailable, error states
Font Family	Helvetica Neue / Arial	Primary UI font (system stack)
Border Radius	4px / 8px	Cards (8px), buttons (4px)

10. TESTING

10.1 Automated Test Framework (ATF) – Unit & Integration Tests

Test ID	Test Name	Type	Expected Outcome	Status
ATF-001	Catalog search returns results for valid ISBN	Unit	JSON response with book data	Pass
ATF-002	Book request flow creates RITM record	Integration	sc_req_item record created	Pass
ATF-003	Auto-approval fires for eligible renewal	Integration	State = Approved within 5s	Pass
ATF-004	ILL request routes to manager approval	Integration	Approval task created	Pass
ATF-005	SLA task created on request submission	Unit	task_sla record with correct target	Pass
ATF-006	Email notification sent on state change	Integration	Notification log entry created	Pass
ATF-007	Catalog sync updates availability count	Integration	u_available_qty updated	Pass
ATF-008	Overdue scheduled job marks breached SLA	Unit	SLA state = Breached	Pass
ATF-009	RBAC: student cannot access librarian queue	Security	403 Forbidden returned	Pass
ATF-010	Invalid ISBN rejected at form submission	Validation	Error message displayed	Pass

10.2 Performance Testing – JMeter Results

Load test configuration: 500 virtual users, 30-minute ramp-up, 30-minute steady state.

Scenario	Target	Result	95th Percentile	Status
Portal page load	< 2.0 s	1.34 s	1.89 s	Pass
Catalog search API	< 1.5 s	0.98 s	1.31 s	Pass
Book request submission	< 3.0 s	1.87 s	2.45 s	Pass
My Requests page	< 2.0 s	1.12 s	1.67 s	Pass
PA Dashboard load	< 5.0 s	3.21 s	4.10 s	Pass
Concurrent users	500	650 achieved	N/A	Pass

Scenario	Target	Result	95th Percentile	Status
Error rate under load	< 1%	0.3%	N/A	Pass

10.3 Running Tests

```
# Run ATF test suites in ServiceNow:
# Navigate to: Automated Test Framework > Test Suites
# Select: 'Smart Library - Full Regression Suite'
# Click: Run Test Suite > Choose environment: Current instance

# Run JMeter performance tests:
cd tests/performance/
jmeter -n -t smart_library_load_test.jmx \
-l results/load_test_$(date +%Y%m%d).jtl \
-e -o results/report_$(date +%Y%m%d)/

# View HTML report:
open results/report_$(date +%Y%m%d)/index.html
```

11. SCREENSHOTS / DEMO

Screenshots are to be inserted by the team before final submission. The following describes each screen with placeholder boxes.

ID	Screen Name	Description	Screenshot
SS-01	Library Portal Home Page	Shows search bar, category tiles, quick stats, and alert banner.	[Insert PNG]
SS-02	Catalog Search Results	Grid view with availability badges and one-click request button.	[Insert PNG]
SS-03	Book Request Form	Dynamic form with ISBN lookup and real-time field validation.	[Insert PNG]
SS-04	Request Confirmation Screen	Success dialog with request number and expected date.	[Insert PNG]
SS-05	My Requests Tracker	Timeline of all user requests with status color-coding.	[Insert PNG]
SS-06	Flow Designer – Book Request Workflow	Visual workflow diagram with state machine nodes.	[Insert PNG]
SS-07	Librarian Fulfillment Dashboard	Queue with SLA traffic-light indicators and bulk actions.	[Insert PNG]
SS-08	Performance Analytics Dashboard	KPI tiles, bar chart (monthly volume), pie chart (status).	[Insert PNG]

11.1 Demo Walkthrough Flow

Step 01: Open portal URL → Login with student credentials (LDAP SSO).

Step 02: Search for 'Clean Code' in catalog search bar.

Step 03: View book detail → confirm availability (3 copies).

Step 04: Click 'Request This Book' → fill form → submit.

Step 05: View confirmation screen with request REQ0001234.

Step 06: Switch to librarian account → view task in fulfillment queue.

Step 07: Mark task as 'Ready for Pickup' → student receives SMS alert.

Step 08: Student collects book → librarian marks 'Collected'.

Step 09: 14 days later: auto-renewal reminder email sent.

Step 10: Student renews via portal → due date extended by 14 days.

Step 11: Book returned → record closed → analytics updated.

12. KNOWN ISSUES

Issue ID	Title	Severity	Description	Workaround	Status
BUG-001	Catalog sync delay on first load	Medium	LMS catalog takes up to 90s to sync on first daily call	Trigger manual sync from Scheduled Jobs	Open
BUG-002	ILL approval email missing attachments	Low	PDF summary not attaching to ILL approval notification email	Download PDF manually from portal	Open
BUG-003	Mobile: datepicker overflow on small screens	Low	Date picker widget overflows viewport on screens < 360px wide	Use landscape orientation on small devices	In Progress
BUG-004	Renewal not updating LMS due date	High	Due date updated in ServiceNow but not propagated to Koha LMS	Librarian updates LMS manually after renewal	In Progress
BUG-005	PA dashboard slow with 12+ months data	Medium	Dashboard load > 5s when date range exceeds 12 months	Limit dashboard range to 6 months	Open
BUG-006	Duplicate notifications on SLA breach	Low	SLA breach triggers email twice when escalation also fires	Suppress duplicate notification rule pending	Open

Issue Reporting: New bugs should be logged at: <https://github.com/your-org/smart-library-servicenow/issues> using the provided bug report template. Include ServiceNow instance version, browser, user role, and steps to reproduce.

13. FUTURE ENHANCEMENTS

■ AI Book Recommendation Engine	Target: v2.0	Integrate ML model to suggest books based on borrow history, course enrollment, and reading patterns. Use ServiceNow Predictive Intelligence framework.
■ Native Mobile App	Target: v2.0	ServiceNow Mobile Studio app for iOS/Android with push notifications, QR code book scanning, and offline reading list management.
■ RFID / IoT Integration	Target: v2.1	Connect to RFID-enabled shelving for automatic check-in/check-out detection, smart return kiosk support, and real-time shelf occupancy tracking.
■ Virtual Agent Chatbot	Target: v2.1	Deploy ServiceNow Virtual Agent on portal to handle catalog queries, process renewals via NLP, and provide instant support — reducing human desk interactions.
■ Multi-Institution Federation	Target: v3.0	Enable federated ILL network connecting multiple universities on a shared ServiceNow catalog, streamlining cross-institution resource sharing.
■ Predictive Analytics	Target: v2.2	Use PA forecasting to predict high-demand periods, optimize procurement, and proactively manage collection gaps before academic semesters.
■ E-Resource Integration	Target: v2.1	Extend to manage e-book/e-journal subscriptions with SSO access to JSTOR, Springer, ProQuest — unified physical + digital library portal.
■ Accessibility Compliance	Target: v2.0	Achieve WCAG 2.1 AA across all portal widgets with screen-reader optimization, high-contrast theme, and regional language localisation.

Project Team Sign-off

This documentation has been reviewed and approved by all team members:

Name	Role	Signature	Date
SANTHOSHKUMAR. R	Team Lead / Backend Dev	_____	09 Apr 2026
DURGA. R	Frontend Developer	_____	09 Apr 2026
MADHUSHINI. R	Database & Integration	_____	09 Apr 2026
ROHINI. M	Testing & Documentation	_____	09 Apr 2026